UNIT - 1

Introduction to NDT & Radiographic Testing

1. Introduction to Non-Destructive Testing (NDT)

* **NDT** = examining a material, component or weld for defects / properties ***without damaging it or impairing its future usefulness*** .
* Also called **NDE** (Non-Destructive Evaluation) or **NDI** (Non-Destructive Inspection).
* The part is tested and then goes back into service - unlike tensile, impact or hardness tests which destroy / damage the sample.
* Used for: detecting flaws, measuring thickness, finding leaks, checking material properties, quality control.

Destructive vs Non-Destructive Testing

Point	Destructive Test	Non-Destructive Test
Sample	Damaged or broken	Remains usable
Testing done on	Sample / specimen from lot	100% of components possible
Cost	High (loss of material)	Low to moderate
Examples	Tensile, impact, fatigue, bend	RT, UT, MPT, LPT, ECT
Information	Actual strength values	Presence, size, position of flaws
Repeatability	Cannot repeat on same piece	Can be repeated many times

Objectives / Need of NDT

* Ensure ***safety and reliability*** of the product (boilers, aircraft, bridges, pipelines).
* Detect defects in early stage and ***reduce cost of rejection*** .
* Prevent accidents and failure in service.
* Maintain uniform quality, support process control.
* Helps ***life extension*** of old plants by in-service inspection.
* Required by codes and standards (ASME, API, IS, ISO).

Types of Defects

* **Surface defects** - cracks, laps, seams, pits, undercut, spatter.
* **Sub-surface defects** - just below the surface (porosity, inclusion).
* **Internal / volumetric defects** - blowholes, shrinkage cavity, slag, voids.
* **Planar defects** are more dangerous than volumetric ones (sharp, stress raiser).

Common NDT Methods

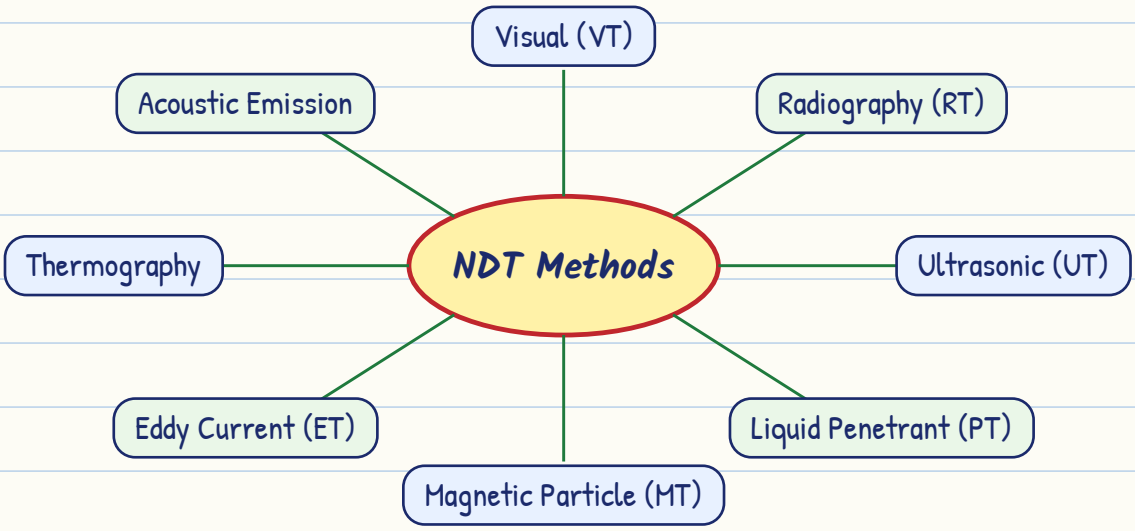


Fig: Principal NDT methods

* **Surface methods:** VT, PT, MT (and ET for near-surface).
* **Volumetric methods:** RT and UT detect internal flaws.

> **Remember:**
> Every NDT method has a **limited range of flaws it can detect** - so often two methods are combined (e.g. UT + MT, RT + PT) for complete inspection.

2. Radiographic Testing (RT) - Principle

* RT uses **short wavelength electromagnetic radiation** (X-rays or Gamma rays) that can penetrate opaque materials.
* Radiation passes through the specimen; **different parts absorb different amounts** .
* Thin section / void / crack absorbs **less** -> more radiation reaches film -> **darker** image.
* Thick section / dense inclusion absorbs **more** -> **lighter** image on film.

* The image recorded on film (radiograph) is a **shadow picture** of internal structure.
* Intensity after absorption: **I = I0 e^(-mu x)** where mu = linear attenuation coefficient, x = thickness, I0 = incident intensity.

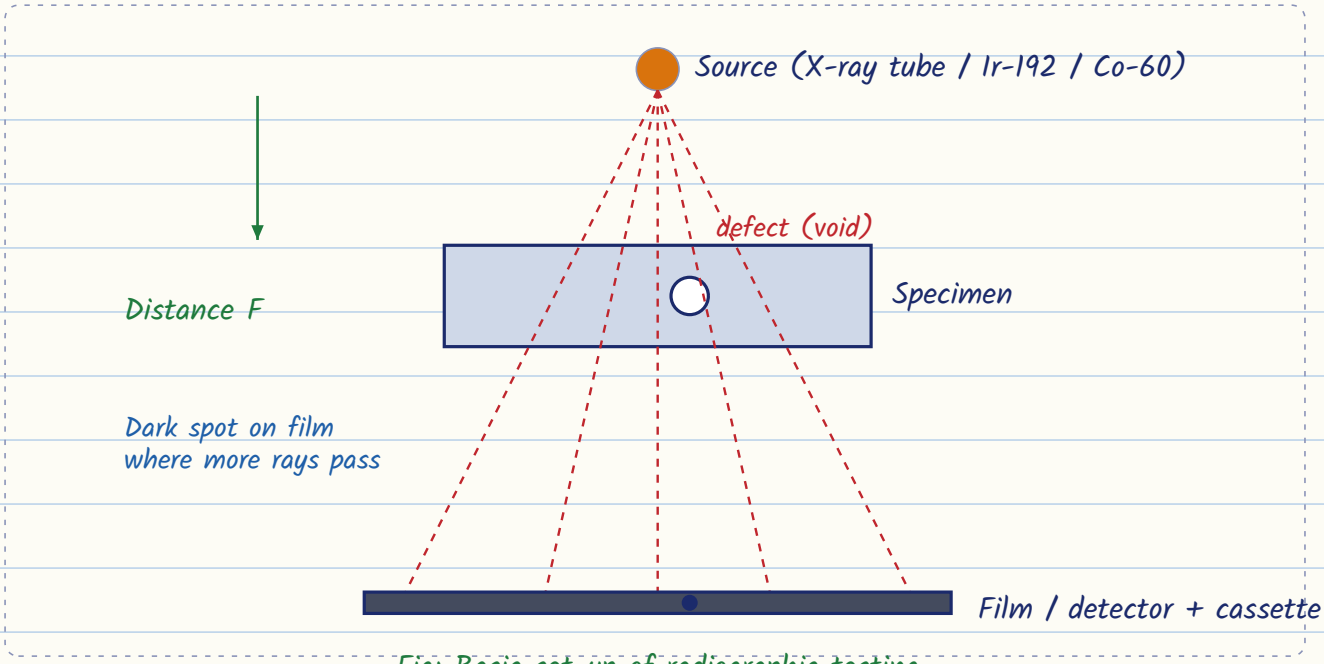


Fig: Basic set-up of radiographic testing

Steps in a Radiographic Test

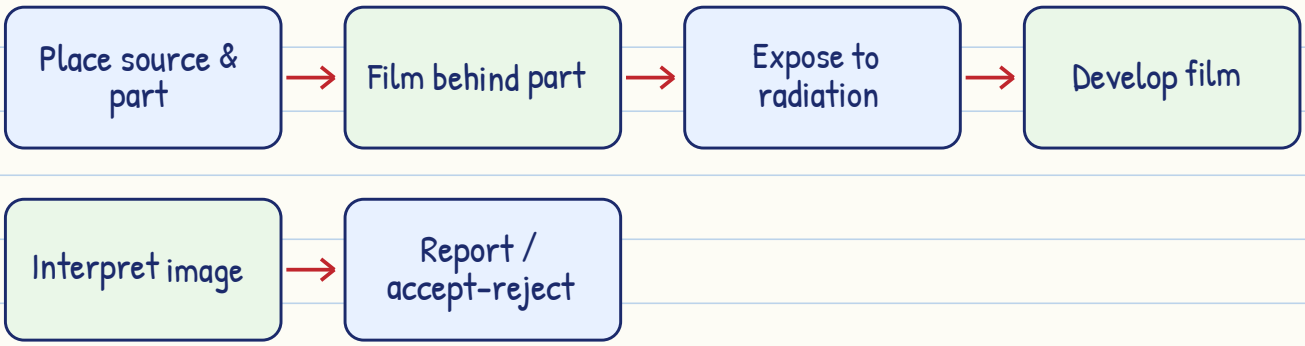


Fig: Radiographic procedure

* Film is loaded in a light-tight **cassette** with lead screens, placed on the side opposite to the source.
* **Penetrameter (IQI)** is kept on the source side to prove the sensitivity of the radiograph.

3. Electromagnetic Radiation – Basics

* X-rays and gamma rays are part of the electromagnetic spectrum; wavelength about **0.001 nm to 10 nm** .

* Travel with speed of light, in straight lines, cannot be deflected by electric or magnetic field.
* They are **invisible, ionising** , affect photographic film and cause fluorescence.
* Energy of a photon: $E = h f = h c / \lambda$ (h = Planck constant).
* Shorter wavelength = higher energy = more penetrating power.
* Quantity measured in keV or MeV; **hard X-rays** (short wavelength) penetrate more, **soft X-rays** penetrate less.

4. Sources of X-rays

4.1 Production of X-rays

* X-rays are produced when **high-speed electrons are suddenly stopped** by a heavy metal target.
* About **99% of energy becomes heat** and only about 1% becomes X-rays - hence cooling is a must.

4.2 X-ray Tube (Coolidge tube)

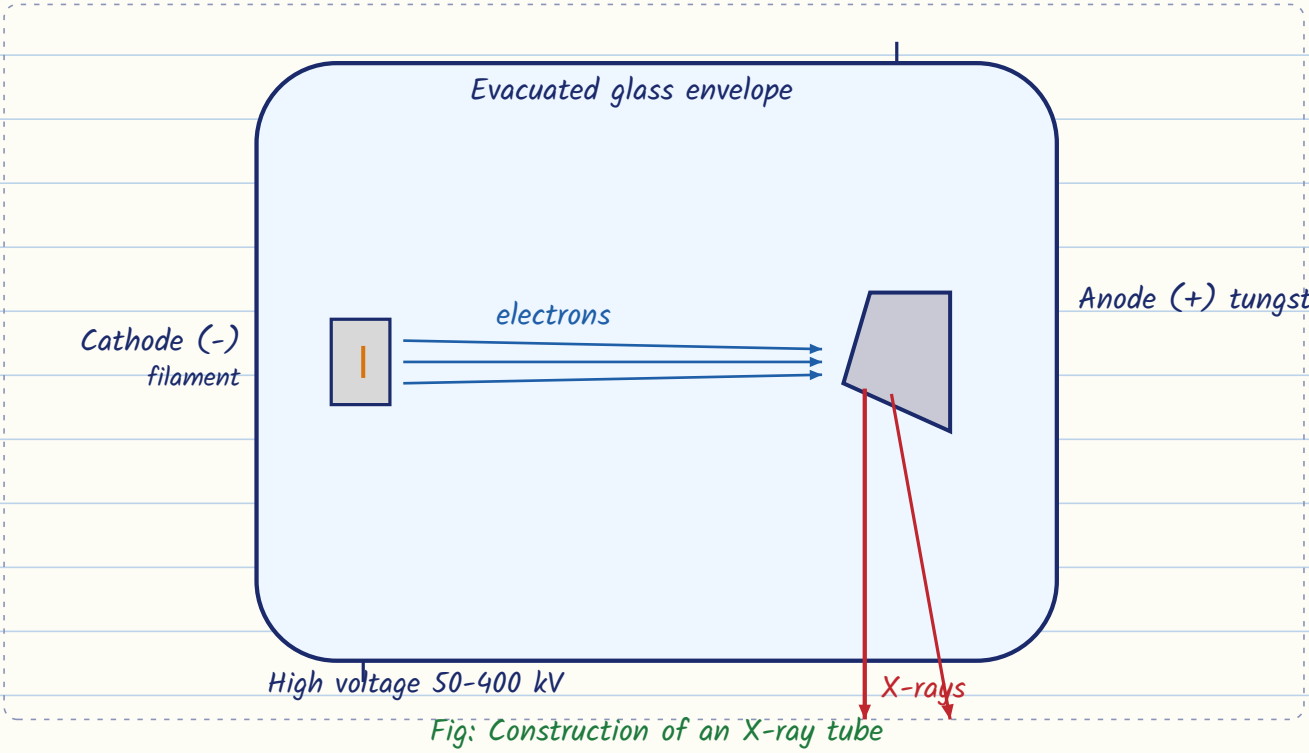


Fig: Construction of an X-ray tube

* **Cathode:** tungsten filament heated by low voltage current; emits electrons by thermionic emission. A focusing cup directs the electrons.
* **Anode:** tungsten (high melting point) target embedded in copper block (good heat conductor).

* **High voltage (kV)** between cathode and anode accelerates the electrons - kV controls **penetration** (quality).
* **Tube current (mA)** = filament heating - controls **quantity / intensity** of X-rays.
* **Vacuum envelope** of glass / metal prevents oxidation of filament and collisions of electrons with gas.
* **Window** of beryllium / glass allows X-rays to leave the tube.
* **Focal spot:** area of target bombarded by electrons - small focal spot gives sharper image.

4.3 Types of X-ray Spectrum

* **Continuous (Bremsstrahlung) spectrum:** electrons slowed in the field of nucleus lose energy as radiation of all wavelengths above a minimum value.
* **Characteristic spectrum:** an electron knocks out an inner shell electron of the target atom; the outer electron fills the gap and emits radiation of definite wavelength.
* Minimum wavelength: **lambda min = 12.4 / V (kV)** in Angstrom.

Key point:

Increase kV -> shorter wavelength, more penetration, lower contrast. Increase mA -> more intensity, shorter exposure time.

4.4 Other X-ray Generators

* **Portable X-ray units** for field work (up to 300 kV), tank-type oil-cooled.
* **Linear accelerator (LINAC):** accelerates electrons up to 4 - 25 MeV for very thick sections.
* **Betatron:** electrons accelerated in circular orbit by changing magnetic field; 10 - 30 MeV, used for steel up to 500 mm.
* **Van de Graaff generator:** electrostatic accelerator, 1 - 4 MeV.
* **Micro-focus X-ray units:** focal spot of a few microns for small electronic parts.

5. Sources of Gamma Rays

* Gamma rays are emitted from the **nucleus of unstable (radioactive) isotopes** during decay.
* Emission energy is fixed and characteristic of the isotope (monochromatic).
* Natural isotopes (radium, radon) were used earlier - now **artificial isotopes** made in nuclear reactors by neutron bombardment.
* **Activity** is measured in Curie (Ci) or Becquerel (Bq): 1 Ci = 3.7 x 10^10 Bq.
* **Half-life:** time for activity to fall to half of its value. Activity: **A = A0 e^(-0.693 t/T)** .
* **Specific activity:** activity per unit mass (Ci per gram) - high value gives small source size and sharper image.

Common Industrial Isotopes

Isotope	Half-life	Energy (MeV)	Steel thickness
Iridium-192 (Ir-192)	74 days	0.31 - 0.61	10 - 75 mm
Cobalt-60 (Co-60)	5.3 years	1.17 and 1.33	50 - 150 mm
Cesium-137 (Cs-137)	30 years	0.66	20 - 90 mm
Thulium-170 (Tm-170)	128 days	0.084	up to 10 mm (thin parts)
Selenium-75 (Se-75)	120 days	0.12 - 0.40	5 - 30 mm

X-ray vs Gamma Ray Source

Point	X-ray	Gamma ray
Origin	Electron stopping in target	Nucleus decay of isotope
Energy	Variable (kV adjustable)	Fixed for the isotope
Power supply	Needed	Not required (portable)
Switch off	Possible	Cannot be switched off
Intensity	High, controllable	Low, decays with time
Cost	High equipment	Low, but source replaced
Image quality	Better contrast	Slightly lower contrast

6. Interaction of X and Gamma Rays with Matter

* When radiation passes through matter it loses energy by **absorption and scattering**, the beam is **attenuated**.
* Narrow beam: **$I = I_0 e^{(-\mu x)}$**; Half Value Layer (HVL) = 0.693 / mu – thickness that reduces intensity to half.

6.1 Photoelectric Effect

* Photon is **completely absorbed** by an inner-shell electron, which is ejected as a photoelectron.
* Dominant at **low energy** (below about 0.1 MeV) and in high atomic number materials (Z^3 dependence).
* Gives high contrast in radiograph.

6.2 Compton Scattering

* Photon strikes a loosely bound outer electron, gives up part of its energy and is **scattered in a new direction** with longer wavelength.
* Dominant in the **medium energy range** (0.1 – 5 MeV); nearly independent of Z.
* Scattered radiation spoils image quality (fog), so lead screens and masks are used.

6.3 Pair Production

* Photon of energy more than **1.02 MeV** passes near a nucleus and converts into an **electron-positron pair**.
* Becomes important above about 5 MeV (high-energy LINAC / betatron).

6.4 Rayleigh (Coherent) Scattering

* Low energy photons are scattered without energy loss; minor effect in radiography.



Fig: Dominant interaction vs photon energy

Remember:

Total attenuation coefficient mu = tau (photoelectric) + sigma (Compton) + kappa (pair production).

7. Radiographic Equipment

7.1 Radiation Sources

* X-ray machines (portable 100 - 300 kV, fixed 400 kV, MeV units) and isotope projectors (exposure devices / cameras).

7.2 Gamma Ray Projector (Exposure Device)

* Heavy shielding container of depleted uranium / lead with an **S-shaped tube** inside which the source capsule is stored.
* Source is pushed out with a **drive cable and crank** through a guide tube to the collimator position, then retracted.
* Has a lock, warning label, and safety interlock.

7.3 Film and Cassettes

* **Industrial X-ray film:** silver bromide emulsion coated on both sides of a transparent base.
* **Film speed:** fast film = coarse grains, short exposure, poor definition; slow film = fine grain, long exposure, better sharpness.
* **Intensifying screens:** lead foil screens (front 0.05 - 0.15 mm, back 0.1 - 0.25 mm) increase blackening by emitting electrons and absorb scattered radiation. Fluorescent screens (calcium tungstate) used in medical type.
* **Cassette:** light tight flexible holder pressing film and screens together.

7.4 Film Processing

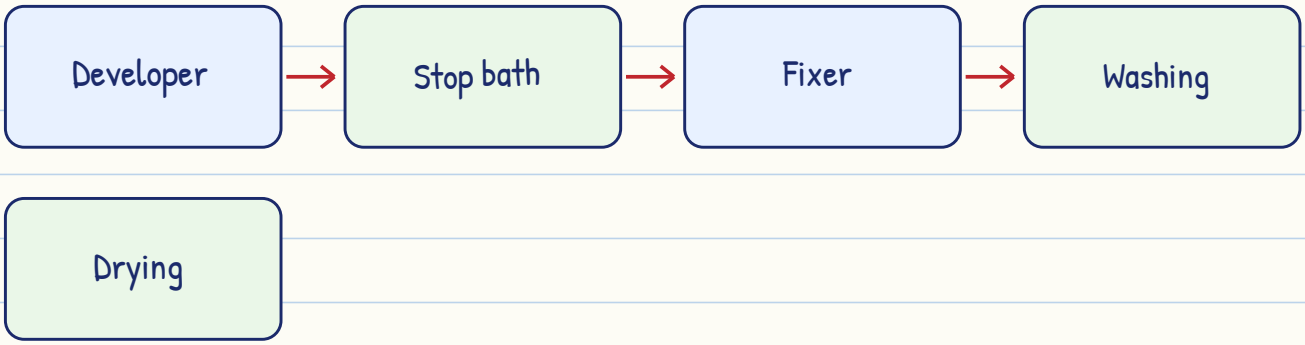


Fig: Darkroom processing sequence

* Developer converts exposed silver halide into black metallic silver; stop bath halts development; fixer removes unexposed silver bromide; washing removes chemicals.

7.5 Radiographic Image Quality

* **Radiographic density (blackening)** $D = \log10 (10 / It)$ - the usual range for acceptance is **1.8 to 4.0**.
* **Contrast:** difference in density between two areas; depends on subject contrast and film contrast.
* **Definition (sharpness):** clearness of edges; **unsharpness** $Ug = F \times t / d$ (F = focal spot size, t = object to film distance, d = source to object distance).
* **Sensitivity:** smallest flaw detectable - checked with **IQI / penetrameter** (wire type or hole type). Sensitivity = (smallest visible step / thickness) x 100 (%).

7.6 Other Detection Systems

* **Fluoroscopy / Real-time radiography:** fluorescent screen with image intensifier - instant view, used for moving parts.
* **Digital radiography (DR, CR):** imaging plates or flat panel detectors, image on computer, no chemicals.
* **Computed tomography (CT):** cross-section slices of part by many projections.
* **Xeroradiography:** selenium plate with electrostatic image.

8. Radiographic Techniques

8.1 Single Wall Single Image (SWSI)

* Source on one side, film on the other, only one wall of object is viewed. Best technique for plates and large pipes; source outside / inside.

8.2 Double Wall Single Image (DWSI)

* Both walls are penetrated but only the **wall near the film is sharply imaged** (source close to the pipe). Used for small bore pipes more than 90 mm.

8.3 Double Wall Double Image (DWDI)

* Both walls appear on the film as an **ellipse** (source offset) - for pipes less than 90 mm outer diameter. Two exposures at 90 degrees are taken.

8.4 Panoramic Technique

* Source at the centre of a cylinder / sphere; film wrapped all around – *full circumferential weld in one exposure* and saves time.

8.5 Other Techniques

* *Stereoradiography:* two shifts of source give 3-D depth location of flaws.
* *Flash / Real-time radiography* for production lines.
* *Source to film distance (SFD):* kept so that geometric unsharpness is limited; increase SFD for better sharpness (inverse square law).
* *Inverse square law:* I1 / I2 = (d2 / d1)^2 – intensity falls with square of distance.
* *Exposure time:* decided by exposure chart (thickness, kV, mA, distance) or isotope exposure calculator (activity, thickness).

Radiography rule:

Exposure = Intensity (mA or Ci) x Time. Reduce kV for contrast, increase distance for sharpness, use lead screens to cut scatter.

8.6 Interpretation of Radiographs

Flaw	Appearance on film
Porosity	Round dark spots (single or cluster)
Slag inclusion	Irregular dark lines or blobs
Crack	Fine, dark, jagged line – most serious
Lack of fusion	Straight dark line along weld edge
Incomplete penetration	Dark line at root centre
Undercut	Dark line along the toe
Tungsten inclusion	Bright (white) spot – denser than steel

* Viewing is done in a *darkened room with a high intensity film viewer (illuminator)* ; acceptance as per code (ASME Sec. V, IS 2595, etc.).

Advantages & Limitations of Radiography

* *Advantages:* detects internal flaws; gives *permanent record* ; suitable for metals and non-metals; shows nature of flaw.

* **Limitations:** planar flaws (cracks) at wrong angle can be missed; **radiation hazard**; costly; needs access to both sides; thick sections need high energy; slow.

9. Safety Aspects of Industrial Radiography

* Ionising radiation causes **biological damage** to cells - may cause burns, cataract, cancer, genetic effects. Effects are cumulative.

9.1 Units of Radiation

Quantity	Unit
Activity	Becquerel (Bq), Curie (Ci)
Exposure	Roentgen (R), C/kg
Absorbed dose	Gray (Gy) = 1 J/kg; rad (100 rad = 1 Gy)
Equivalent / effective dose	Sievert (Sv); rem (100 rem = 1 Sv)

9.2 Dose Limits (ICRP / AERB)

* Radiation worker: **20 mSv per year averaged over 5 years (max 30 mSv in any one year)**.
* Members of the public: **1 mSv per year**.
* Pregnant worker: abdomen dose limit 1 mSv for the rest of pregnancy.

9.3 Three Basic Principles - TDS



Fig: Time - Distance - Shielding

* **Time:** dose is proportional to time of exposure.
* **Distance:** dose falls with the square of distance (inverse square law).
* **Shielding:** lead, steel, concrete, depleted uranium absorb radiation. Concrete about 15 cm per HVL-based design; lead for portable shielding.

9.4 Safety Measures

* Establish **controlled and supervised area** with barriers, rope and **radiation warning signs (trefoil)**, red light / alarm during exposure.
* Personnel monitoring: **film badge, TLD (thermoluminescent dosimeter), pocket dosimeter**; direct reading alarm ratemeter.

* Area survey with **Geiger-Muller counter / survey meter** before and after exposure.
* Only **trained and licensed** radiographers (AERB certification in India); regular medical check-ups.
* Proper storage of sources in locked bunkers; **source leak test** and record keeping; transport as per rules.
* Emergency procedure for **stuck source** - evacuate, cordon, inform Radiation Safety Officer (RSO).
* Use of **collimators and lead masks** to limit beam; use of enclosed **exposure rooms with interlocks** .

Summary:

RT = best for internal flaws and permanent record; X-ray for controllable, gamma for portable; interaction by photoelectric, Compton, pair production; always follow TDS and monitoring for safety.

Important Short Questions

* Define NDT. List its advantages.
* Differentiate X-rays and gamma rays.
* Explain Compton scattering and pair production.
* What is half-life? Name any four isotopes used in RT.
* State the three principles of radiation protection.